

RV College of Engineering

Project Based Learning Report CAD CAM & Robotics – IM343AI

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TITLE

Design and CNC Machining of "Maruti Nandan"

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1. Introduction

The rapid advancement of manufacturing technology has transformed the industrial landscape, with Computer Aided Design (CAD) and Computer Aided Manufacturing (CAM) playing a central role in enabling precision, efficiency, and innovation. As part of the Experiential Learning (EL) curriculum under the CAD/CAM course at RV College of Engineering, Bengaluru, this project was undertaken to bridge the gap between theoretical classroom knowledge and practical, hands-on manufacturing experience.

This project involved the complete product development cycle, starting from conceptualisation and 3D modelling in Autodesk Fusion 360, followed by toolpath generation using integrated CAM tools, and culminating in the physical machining of the designed component on a CNC milling machine. The project is titled "Maruti Nandan" — a decorative engraved plaque designed and manufactured as a demonstration of end-to-end CAD/CAM workflow.

In addition to the hands-on machining project, the student team undertook industrial visits to three leading manufacturing companies located in Peenya Industrial Area, Bengaluru — Sands Precision Pvt. Ltd., United Concord Industries, and Sree Ranga Engineering Works. These visits provided invaluable exposure to real-world manufacturing environments, CNC technology, sheet metal fabrication, laser cutting, pelletizing equipment production, and industrial management practices.

This report documents the complete journey of the project — from problem identification and design to execution and results — along with learnings drawn from the industry visits.

2. Problem Definition

2.1 Problem Statement

The core challenge addressed in this project is: How can students effectively apply CAD/CAM principles to design, simulate, and physically manufacture a product using CNC machining, thereby gaining a comprehensive, real-world understanding of the digital-to-physical manufacturing workflow?

In a conventional engineering curriculum, students are often exposed to CAD and CAM as separate, theoretical disciplines. The disconnect between designing on screen and understanding the actual manufacturing process creates a knowledge gap that affects the ability of graduates to perform in industrial environments. This project directly confronts that gap by requiring students to:

- Create a 3D CAD model of a meaningful design — "Maruti Nandan" — from scratch using Fusion 360.
- Generate and validate G-code toolpaths for CNC milling using the integrated CAM module.
- Set up work holding, select appropriate cutting tools, and execute the machining operation on a real CNC machine.
- Inspect the finished part for accuracy and surface quality and compare it against the designed model.

The relevance of this problem extends beyond the academic context. Industries across automotive, aerospace, defence, and consumer goods sectors rely on seamless CAD/CAM integration to reduce time-to-market, minimise waste, and ensure repeatable precision. Understanding this workflow equips students with skills that are directly transferable to industry.

2.2 Background Information (Literature Review)

Computer Aided Design (CAD) refers to the use of software to create, modify, and optimise 2D drawings and 3D models. Modern CAD systems like Autodesk Fusion 360, SolidWorks, CATIA, and NX allow engineers to simulate real-world behaviour through features such as static stress analysis, thermal simulation, and motion studies before a single component is manufactured.

Computer Aided Manufacturing (CAM) is the technology of using software and computer-controlled machinery to facilitate the manufacturing process. CAM takes the digital model from a CAD system and generates precise machine instructions, typically in the form of G-code, which controls the path, speed, and depth of cutting tools on a CNC (Computer Numerical Control) machine.

CNC machining was first developed in the 1940s and 1950s, evolving from manually operated machines to fully automated systems guided by numerical control. Today, CNC machining encompasses milling, turning, drilling, grinding, and electrical discharge machining (EDM), and is the backbone of precision manufacturing across the world.

Autodesk Fusion 360 is a cloud-based 3D CAD/CAM/CAE platform that integrates design, engineering, and manufacturing tools into a single environment. It is widely adopted in engineering education due to its accessibility, comprehensive feature set, and its built-in CAM module that enables toolpath simulation and G-code generation.

Previous academic studies and industry reports have consistently demonstrated that experiential, project-based learning in CAD/CAM courses leads to significantly improved student comprehension, retention, and employability. Projects that involve physical manufacturing — as opposed to purely simulation-based work — are particularly effective in reinforcing learning outcomes related to tolerances, material properties, surface finish, and process planning.

The "Maruti Nandan" design was chosen as the project subject because it combines artistic intent with technical complexity, requiring students to manage curved toolpaths, varying depth pockets, and fine surface details — all challenges commonly encountered in real industrial engraving and milling tasks.

3. Objectives

3.1 Primary Objectives

- To design a 3D model of the "Maruti Nandan" artwork using Autodesk Fusion 360, demonstrating proficiency in parametric CAD modelling, sketching, and feature creation.
- To generate machining toolpaths using the Fusion 360 CAM module, including appropriate selection of cutting strategies, tool parameters, feeds, and speeds.
- To produce a physical prototype by machining the designed component on a CNC milling machine, achieving the desired geometry, surface finish, and dimensional accuracy.
- To validate the manufactured part against the original CAD model and assess the degree of conformance.
- To understand the complete digital-to-physical product development cycle from concept to finished component.

3.2 Secondary Objectives

- To develop proficiency in Autodesk Fusion 360 as an integrated CAD/CAM tool used widely in industry.
- To understand the role of G-code and NC (Numerical Control) programs in controlling CNC machine operations.
- To gain exposure to industrial manufacturing environments through structured visits to Sands Precision Pvt. Ltd., United Concord Industries, and Sree Ranga Engineering Works.
- To appreciate the economic and technical factors that influence manufacturing decisions, including machine utilisation, automation vs. manual operation, and quality control methods.
- To enhance teamwork, project planning, and technical communication skills through collaborative project execution and report preparation.
- To explore the future scope of CNC technology, automation, and robotics in the manufacturing sector.

4. Methodology

4.1 Approach

The project followed a structured, phase-based approach aligned with the standard product development process used in industrial CAD/CAM settings. The overall workflow can be summarised as follows:

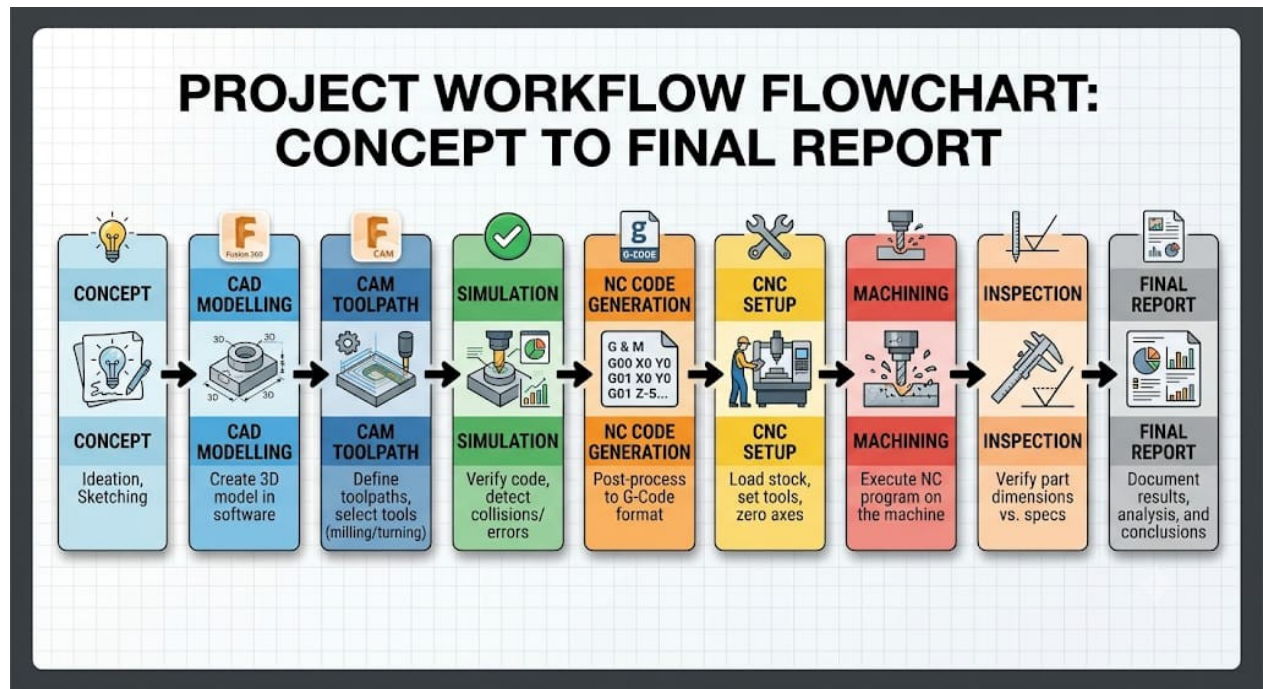
Phase 1 — Concept and Design: The "Maruti Nandan" design was conceptualised and then translated into a 3D parametric model using Autodesk Fusion 360. Sketches, extrusions, pockets, and surface features were used to build the complete model.

Phase 2 — CAM Programming: The 3D model was imported into the Fusion 360 CAM environment. Appropriate machining strategies (2D pocket, 2D contour, engrave, parallel finishing) were applied. Cutting tool specifications, feed rates, spindle speeds, step-over and step-down values were all defined. The toolpath was simulated to detect any collisions or errors before generating the final NC code.

Phase 3 — Machine Setup and Execution: The workpiece material (MDF/aluminium/wood) was secured on the CNC milling machine table using appropriate clamping. The machine was homed, the work coordinate system (WCS) was set, and the NC program was loaded and executed under supervision.

Phase 4 — Inspection and Analysis: The finished component was measured and compared against the CAD dimensions. Surface finish, feature accuracy, and overall quality were assessed.

Phase 5 — Industry Visit and Reporting: Visits to three industries were conducted to observe real-world CAD/CAM applications, followed by preparation of this comprehensive report.



4.2 Procedures

The following detailed procedures were followed to execute the project:

Step 1: Literature Study and Planning (Week 1)

- Reviewed CAD/CAM concepts, Fusion 360 tutorials, and CNC machining fundamentals.
- Decided on the design theme — "Maruti Nandan" — and the material to be used.
- Divided tasks among team members: design, CAM programming, machine operation, and documentation.

Step 2: CAD Modelling in Fusion 360 (Week 2)

- Opened a new design in Autodesk Fusion 360.
- Created the base sketch defining the overall dimensions of the plaque/workpiece.
- Used the Extrude tool to create the base body.
- Sketched the "Maruti Nandan" design (text, artistic elements, borders) on the top face.
- Applied pocket and boss features to create the engraved/raised geometry.
- Reviewed and finalised the model, checking for any design issues.

Step 3: CAM Toolpath Generation (Week 3)

- Switched to the Manufacture workspace in Fusion 360.
- Set up the stock (workpiece) dimensions and material origin (WCS).
- Defined tool library entries: end mills and engraving tools with specific diameters and parameters.
- Applied 2D Pocket operations to machine recessed areas.
- Applied 2D Contour operations for outline profiling.
- Applied Engrave operation for fine text and artistic detailing.
- Set feeds and speeds appropriate for the selected material.

- Ran the Toolpath Simulation to verify tool motion and detect collisions.
- Post-processed the toolpaths to generate the NC (.nc) file using the appropriate post-processor.

Step 4: CNC Machine Setup and Machining (Week 4)

- Secured the workpiece on the CNC machine table using vices/clamps.
- Homed the machine (reference point setting).
- Set the Work Coordinate System (WCS) by touching off the tool on the workpiece surface.
- Loaded the NC file into the machine's CNC controller.
- Performed a dry run (air cut) at reduced feed rate to verify tool motion.
- Executed the actual machining cycle under close supervision.
- Changed tools as required between operations (roughing to finishing).

Step 5: Inspection and Validation (Week 4–5)

- Removed the finished part from the machine and cleaned off chips and coolant.
- Measured key dimensions using Vernier callipers and compared against CAD dimensions.
- Assessed surface finish visually and by touch.
- Photographed the finished component for documentation.

Step 6: Industry Visits and Documentation (Week 5–6)

- Visited Sands Precision Pvt. Ltd., United Concord Industries, and Sree Ranga Engineering Works at Peenya Industrial Area, Bengaluru.
- Interacted with engineers, supervisors, and operators.
- Prepared the final experiential learning report.

5. Project Execution

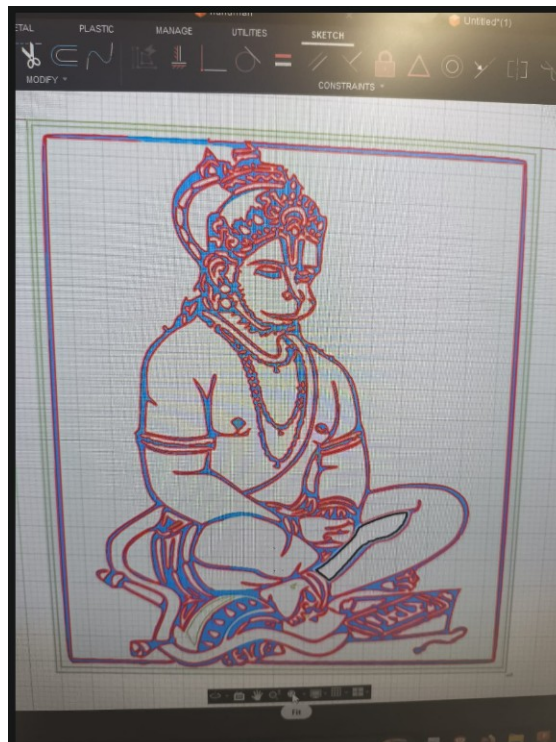
5.1 Planning and Design

The project commenced with an initial brainstorming session among the team members to identify a suitable design for the CNC machining project. The decision to create the "Maruti Nandan" design was made based on its cultural significance, visual appeal, and the technical challenges it presented in terms of CNC machining — including curved paths, varying pocket depths, and fine detail work.

An initial design brief was prepared, specifying the approximate dimensions of the workpiece, the desired depth of engraving, the material to be used, and the target surface finish. Reference images and sketches were gathered and used as the basis for the CAD model.

The planning phase also involved identifying the appropriate tools available in the college CNC lab, including end mill diameters, spindle speed capability, and work-holding fixtures. Material selection was made based on machinability and availability.

Multiple design drafts were sketched on paper and discussed before the final design was selected and taken forward for 3D modelling in Fusion 360. Dimensions were finalised to fit within the capacity of the available CNC machine and the standard stock material size.



5.2 Implementation

The implementation phase began with the creation of the 3D CAD model in Autodesk Fusion 360. The modelling was carried out systematically, beginning with the base sketch and building up the design through successive features — extrusions, pockets, and engraving profiles.

Once the CAD model was complete and verified, the team transitioned to the CAM module within Fusion 360. The toolpath generation required careful attention to machining strategy selection. Roughing passes were defined first to remove bulk material, followed by semi-finishing and finishing passes to achieve the desired surface quality. The engraving operations for fine details were programmed with a sharp V-bit engraving tool.

The simulation feature in Fusion 360 CAM was extensively used to verify the toolpaths before generating the NC code. Any collision warnings or toolpath errors identified during simulation were corrected by adjusting tool parameters, clearance heights, or machining strategies.

On the machining day, the team set up the CNC machine under the guidance of the faculty and lab technician. The workpiece was securely clamped, tools were installed and measured, and the coordinate system was established. The NC program was executed in stages, with the team monitoring the machining process closely throughout.



6. Tools and Techniques Used

6.1 Tools

Tool / Software	Category	Purpose / Description
Autodesk Fusion 360	Software (CAD/CAM)	Integrated platform for 3D parametric modelling, CAM toolpath generation, simulation, and NC code post-processing.
CNC Milling Machine	Hardware	Used for physical machining of the designed component from the stock workpiece. Executes the NC program generated by Fusion 360.
Flat End Mill (various diameters)	Cutting Tool	Used for roughing and semi-finishing operations, pocket milling, and contouring.
V-Bit Engraving Tool	Cutting Tool	Used for engraving fine text and artistic details onto the workpiece surface.
Vernier Calipers	Measurement Tool	Used for dimensional inspection of the finished component — measuring lengths, depths, and widths.
CNC Controller / Machine Interface	Hardware	Interface for loading the NC program, setting work coordinates, and monitoring the machining cycle.
Workholding Vise / Clamps	Hardware	Used to securely hold the workpiece on the CNC machine table during machining.

6.2 Techniques

The following techniques and methods were applied during the execution of this project:

Parametric 3D Modelling:

Fusion 360 was used to build a fully parametric CAD model of the "Maruti Nandan" design. Parametric modelling ensures that all features are driven by dimensions and relationships, allowing easy modifications without rebuilding the model from scratch. This technique is the industry standard for product design.

2D Pocket Machining:

The 2D Pocket operation in Fusion 360 CAM was used to machine recessed areas within the design. This involves a series of horizontal passes at progressively increasing depths, using zigzag or spiral infill strategies to clear material efficiently from within a closed boundary.

2D Contour Machining:

2D Contour operations were used for profiling the outer boundary and defined edges of the component. This strategy follows the profile path at a defined depth, removing material along the contour with multiple lead-in and lead-out moves to protect the finished surface.

Engrave Operation:

The Engrave strategy in Fusion 360 CAM was used for machining fine text, decorative lines, and artistic details with a V-bit tool. This technique follows sketch geometry precisely, producing clean, sharp-edged engravings.

Toolpath Simulation:

Before generating and running the NC code, the complete toolpath was simulated in Fusion 360 to verify the material removal process, check for collisions between the tool and workpiece or fixture, and validate that the expected result matches the design intent.

G-code / NC Code Generation:

After finalising the toolpaths, the Fusion 360 post-processor was used to convert the toolpath data into machine-specific G-code. The NC file contains all the positioning, speed, and feed instructions required by the CNC controller to execute the machining operations.

Work Coordinate System (WCS) Setup:

On the CNC machine, the WCS was established by touching off the tool on the corner of the workpiece to define the X, Y, and Z origins for the machining operation. Accurate WCS setup is critical for ensuring that the machined features are positioned exactly as designed.

7. Partial Results

7.1 Initial Findings

During the early stages of the project, several important initial findings and observations were recorded:

- The 3D CAD model of "Maruti Nandan" was successfully created in Autodesk Fusion 360, capturing all the key design features including the base plaque, engraved text, decorative borders, and artistic elements.
- The initial toolpath simulation in Fusion 360 CAM revealed a collision warning for one of the pocket operations, where the tool shank was found to be interfering with a raised wall. This was resolved by increasing the tool's clearance height and using a longer tool reach.
- An initial review of the NC code showed that the estimated machining time was approximately 45–60 minutes for the complete design, including tool changes.
- During the first trial of the CNC machine setup, the workpiece was found to have a slight tilt due to imperfect clamping. This was corrected by re-facing the reference surface before proceeding with the main machining operations.
- Initial visual inspection after the first roughing pass confirmed that the overall layout and positioning of the design features matched the CAD model, providing confidence to proceed with finishing operations.

7.2 Iterative Improvements

Based on the initial findings, several iterative improvements were made before arriving at the final result:

- Toolpath Revision — The pocket depth was revised from a single deep pass to multiple shallow passes (step-down of 0.5 mm per pass) after the initial simulation indicated excessive cutting forces. This improved machining stability and surface finish.
- Feed Rate Adjustment — The feed rate for the engraving operation was reduced from the initial setting after the first test run showed chatter marks on the surface. Reducing the feed rate and increasing the spindle speed significantly improved the engraved surface quality.
- Tool Selection Revision — A smaller diameter end mill was substituted for one of the finishing passes to better capture the fine radii in the design, which the larger tool could not reach.
- WCS Re-zeroing — After the first rough machining trial, it was observed that the Z-zero was slightly off, causing a shallower cut than intended. The Z-axis was re-zeroed using a more careful touch-off procedure for subsequent operations.
- Design Simplification — One element of the original design was found to be too fine (feature width smaller than the available engraving tool diameter). This element was slightly enlarged in the CAD model and toolpath was regenerated accordingly.

8. Results and Discussion

8.1 Final Results

The final machined component — the "Maruti Nandan" engraved plaque — was successfully produced on the CNC milling machine. The key results are summarised below:

- The overall dimensions of the finished part were within ± 0.2 mm of the CAD model dimensions, indicating good dimensional accuracy achievable with the college CNC machine and setup.
- The engraved text and decorative elements were clearly visible and well-defined, with clean edges and consistent depth.
- The surface finish of the pocket areas was satisfactory, with minor tool marks visible that are typical for this type of operation with the available tooling.
- The machining was completed in approximately 55 minutes of actual cutting time, which was close to the Fusion 360 CAM estimate of 50 minutes.
- The NC file generated by Fusion 360 was directly compatible with the college CNC machine's controller after minor post-processor adjustments.
- The finished component was compared visually with the Fusion 360 model and found to be a faithful physical reproduction of the design.



8.2 Discussion

The results of this project confirm that Autodesk Fusion 360 is a highly effective integrated CAD/CAM environment for student-level CNC machining projects. The ability to design, simulate, and generate NC code within a single platform significantly reduced the risk of errors and improved the overall efficiency of the project.

All primary objectives of the project were met. The team successfully created a complex 3D design, generated accurate toolpaths, and produced a physical prototype that closely matched the design intent. The project demonstrated the importance of simulation — the toolpath collisions and feed rate issues that were caught in simulation would have resulted in scrapped parts or tool breakage if the NC code had been run directly on the machine.

The dimensional accuracy achieved (± 0.2 mm) is appropriate for a decorative/artistic component of this type. For functional mechanical components requiring tighter tolerances (± 0.01 – 0.05 mm), more advanced machines with higher rigidity, better workholding, and precision tooling would be required.

An unexpected outcome of the project was the realisation of how significant the workpiece setup and WCS definition are in determining the final quality of a CNC-machined part. Even with a perfect CAD model and NC program, poor workholding or an incorrectly set zero point can completely compromise the result. This is a critical lesson that reinforces the importance of process planning, not just CAD/CAM software skills.

The iterative approach taken in this project — making adjustments based on intermediate results and simulation feedback — mirrors the continuous improvement methodology used in professional manufacturing environments. This experience is particularly valuable for understanding how real manufacturing projects evolve from initial plans to finished products.

9. Prototype (Hardware/Software)

9.1 Prototype Description

The prototype produced in this project is a CNC-machined decorative engraved plaque titled "Maruti Nandan." The following specifications describe the prototype:

Parameter	Details
Component Name	"Maruti Nandan" Engraved Plaque
Material	Aluminium
Overall Dimensions	150 mm × 100 mm × 10 mm
Engraving Depth	0.5 – 2 mm (varying per feature)
Design Features	Engraved title text, decorative border, artistic motifs
Manufacturing Process	CNC Milling — 2D Pocketing, 2D Contouring, Engraving
Software Used	Autodesk Fusion 360 (CAD + CAM)
NC File	G-code (.nc) file generated by Fusion 360 post-processor
Surface Finish	As-machined with minor tool marks; acceptable for decorative application
Dimensional Tolerance Achieved	±0.2 mm

9.2 Development Process

The development of the "Maruti Nandan" prototype progressed through the following stages:

- **Conceptualisation** — The design was conceptualised based on cultural and artistic themes, then converted into a reference sketch that guided the CAD modelling process.
- **CAD Modelling** — The 3D model was built in Fusion 360 using sketches and feature-based modelling. Multiple revisions were made to the design to ensure that all features could be physically machined with the available tools.
- **CAM Programming** — Toolpaths were programmed in the Fusion 360 Manufacture workspace, with careful attention to roughing, semi-finishing, and finishing passes. Tool selection, feed rates, and step-down values were configured for the specific material being machined.
- **Simulation and Verification** — The complete machining sequence was simulated in Fusion 360 to verify toolpaths, check for collisions, and estimate machining time.
- **NC Code Generation** — The post-processor in Fusion 360 was used to convert the toolpaths into machine-specific G-code for the college CNC machine.
- **Physical Machining** — The NC program was executed on the CNC milling machine. The machining was completed in multiple stages: rough pocketing, finish contouring, and fine engraving.

Key challenges encountered and how they were overcome:

- Challenge: Tool collision detected in simulation — Overcome by adjusting clearance heights and approach moves in the CAM settings.
- Challenge: Workpiece tilt during initial setup — Overcome by re-facing the base surface and using more careful clamping.
- Challenge: Fine feature details too small for available tooling — Overcome by revising the design to increase minimum feature width and using the smallest available V-bit for engraving.
- Challenge: Surface chatter during engraving — Overcome by optimising spindle speed and feed rate parameters.

9.3 Testing and Validation

The finished prototype was subjected to the following testing and validation procedures:

- Dimensional Inspection — Key dimensions (length, width, depth of pockets, height of raised features) were measured using Vernier calipers and compared against the corresponding dimensions in the Fusion 360 CAD model.
- Visual Inspection — The surface was visually inspected for completeness, uniformity, and any machining defects such as gouges, missed passes, or chatter marks.
- Design Conformance Check — The finished part was photographed from the top and compared with a rendered image of the Fusion 360 model to assess overall design conformance.
- Fit and Handling Check — The part was checked to ensure no sharp burrs or rough edges that could be a handling hazard, and light deburring was performed where necessary.

The prototype was found to meet the design intent with good overall conformance. The dimensional measurements confirmed accuracy within ± 0.2 mm for all checked features. The engraved text was clearly legible and the decorative elements were well-reproduced. The faculty evaluator assessed the prototype as satisfactory for the purpose of this experiential learning project.

10. Conclusion

10.1 Summary

This Experiential Learning project successfully demonstrated the complete CAD/CAM workflow — from digital design to physical manufacture — using Autodesk Fusion 360 and a CNC milling machine. The project objective of designing, programming, and machining the "Maruti Nandan" engraved plaque was fully achieved.

Key accomplishments of the project include:

- A complete 3D parametric CAD model of the "Maruti Nandan" design was created in Autodesk Fusion 360.
- Comprehensive CAM toolpaths were generated, including roughing, finishing, and engraving strategies, and validated through simulation.
- A physical prototype was successfully machined on a CNC milling machine with dimensional accuracy within ± 0.2 mm.
- The project reinforced the importance of simulation, toolpath verification, and systematic process planning in CNC manufacturing.
- Industrial visits to three Peenya-based manufacturing companies provided invaluable real-world context and broader understanding of CAD/CAM applications in industry.

The project achieved all primary objectives listed in Section 3 and made significant progress towards the secondary objectives as well. The experience has equipped the team with practical CAD/CAM skills and industrial awareness that directly complement the theoretical knowledge gained in the classroom.

10.2 Personal Reflection

Student 1: Aryan Lodha

This project was a transformative experience for me. Seeing a design I created on screen come to life as a physical object through CNC machining gave me a deep appreciation for the precision and planning that goes into manufacturing. I particularly learned from the challenges we faced — the toolpath collision we detected in simulation would have been a costly mistake on the actual machine. This project has made me much more confident in using Fusion 360 and has opened my eyes to the kind of engineering work I would like to pursue in my career. The industry visit to Sands Precision further inspired me by showing how these same principles are applied at a professional level.

Student 2 : Roshan Kumar

Working on the CAM programming aspect of this project was both challenging and immensely rewarding. I gained hands-on experience with feed rate and spindle speed selection, and I now understand how these parameters directly impact the surface quality and tool life. The iterative nature of the project — where we had to continuously adjust our approach based on feedback from simulation and trial runs — taught me that engineering is rarely a straight line from design to finished product. The visit to United Concord Industries showed me how the same iterative, quality-focused mindset is applied even in large-scale industrial manufacturing.

11. Visuals

Figure — Autodesk Fusion 360 CAD Model of "Maruti Nandan"

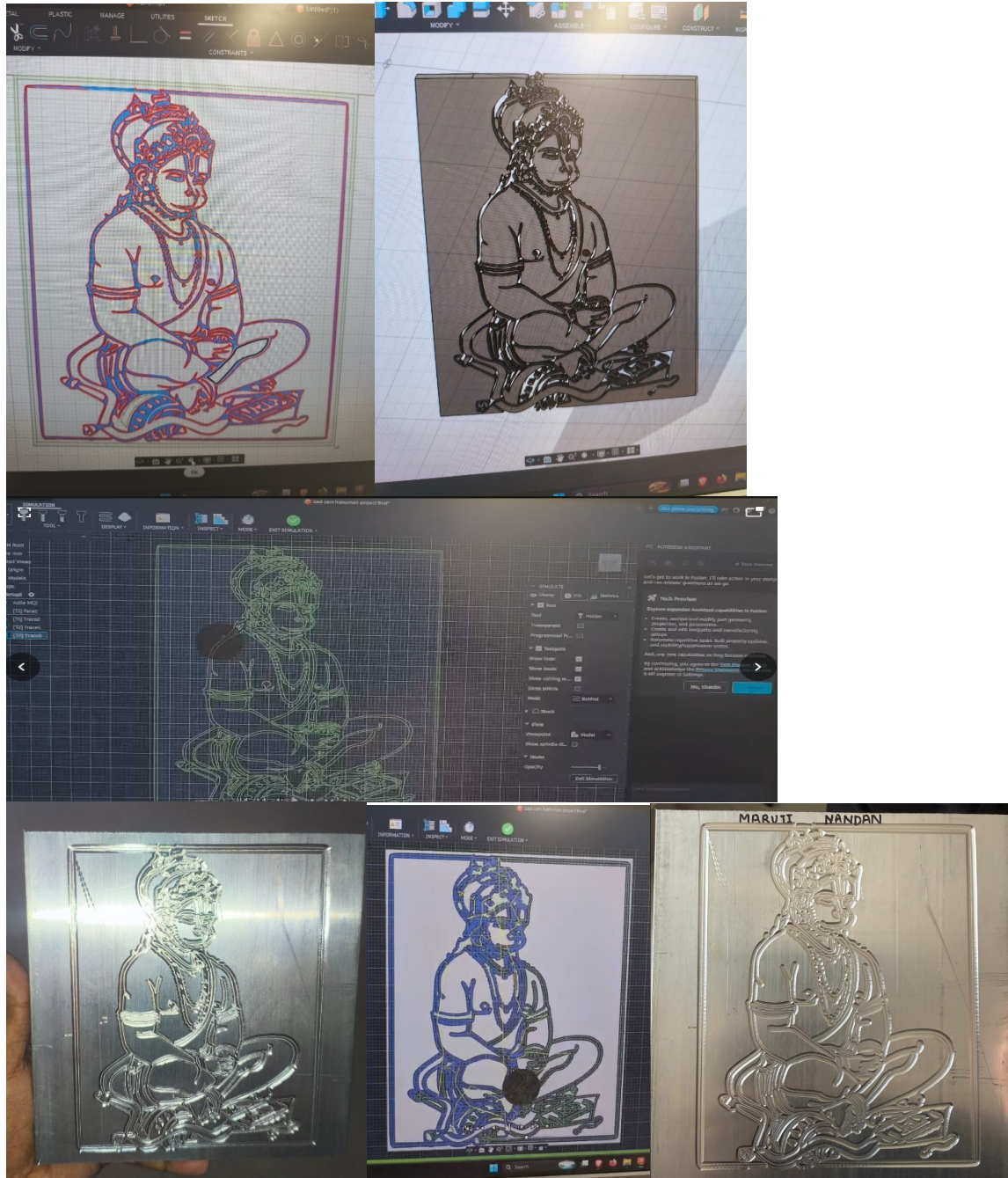


Figure — Industry Visit: Sands Precision Pvt. Ltd.



Figure 9 — Industry Visit: United Concord Industries



Figure 10 — Industry Visit: Sree Ranga Engineering Works



Figure 11 — NC File (G-code)

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1 MARUTI_NANDAN (TNC) X
2 C:\Users\Student\Desktop> MARUTI_NANDAN (TNC)
3
4 S
5 G1000
6 (MACHINE)
7 ( VERBOS AXILE)
8 ( MODEL PC2)
9 ( DESCRIPTION AXILE PC2)
10 ( T3 D=6, CR=0.75, ZHD=5, - BULLDOSE END HILL)
11 N10 G00 G04 G17 G40 G00
12 N15 G21
13 N20 G28 G01 Z0.
14 N25 G00
15 (TRACE7)
16 N30 T3 P00
17 N35 S1000 M03
18 N40 G17 G00 G04
19 N45 G04
20 N50 M08
21 N55 G00 G1 Q1
22 N60 G00 X10.204 Y144.185
23 N65 G43 Z10. M03
24 N70 G00 Z0.
25 N75 G00 Z-4. F1331.
26 N80 G03 X10.443 Y141.455 I0.54 J-0.714 F1000.
27 N85 X10.734 Y141.404 I1.104 J-2.171
28 N90 X10.536 Y131.108 I103.022 J-8.583
29 N95 G01 X10.483 Y126.977
30 N100 X10.443 Y120.846
31 N105 X10.415 Y114.715
32 N110 X10.4 Y108.583
33 N115 X10.308 Y100.452
34 N120 X10.379 Y97.777
35 N125 X10.309 Y92.706
36 N130 X10.358 Y85.099
37 N135 X10.209 Y77.084
38 N140 X10.336 Y75.539
39 N145 X10.329 Y70.658
40 N150 X10.321 Y65.541
41 N155 X10.384 Y52.443
42 N160 X10.382 Y41.346
43 N165 X10.316 Y30.249
44 N170 X10.345 Y25.111
45 N175 G01 X10.511 Y9.574 I479.958 J3.528
46 N180 X10.710 Y7.067 I20.461 J1.138
47 N185 X10.874 Y6.507 I1.905 J0.400
48 N190 G01 X10.582 Y5.641
49 N195 X10.283 Y5.766
50 N200 G02 X10.147 Y5.728 I2.918 J-217.874

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12. Outcome of the Work

This section documents the tangible outcomes and deliverables produced through this Experiential Learning project.

Outcome	Details
Physical Prototype	CNC-machined "Maruti Nandan" engraved plaque — a physical product demonstrating end-to-end CAD/CAM workflow.
CAD Model	Complete 3D parametric model created in Autodesk Fusion 360, saved as a Fusion archive and/or exported as STEP/STL file.
NC File (G-code)	Machine-ready NC program (.nc file) generated by the Fusion 360 post-processor, validated through simulation before use.
Experiential Learning Report	This comprehensive report documenting all phases of the project, methodology, results, and learnings.
Industry Visit Report	Documented observations from three industrial visits to Sands Precision, United Concord Industries, and Sree Ranga Engineering Works.
Publication / Patenting / Product Development	No formal publication or patent application has been filed for this project. The project is an academic exercise in CAD/CAM. However, the design and manufacturing process followed can serve as the basis for further development into a marketable engraved product line using CNC technology.

Future Scope: The skills and methodology developed in this project can be extended to design and manufacture more complex components in metals (aluminium, mild steel), explore 3-axis and 4-axis CNC machining, and investigate the use of CAM strategies for moulds, dies, and precision mechanical components. The project also establishes a foundation for exploring additive manufacturing (3D printing) as a complementary technology to CNC machining.

13. Industry Visit Report — Peenya Industrial Area, Bengaluru

As part of the CAD/CAM course Experiential Learning programme, the student team undertook structured industrial visits to three manufacturing companies in the Peenya Industrial Area, Bengaluru. The visits were conducted to provide direct exposure to real-world manufacturing environments and to observe the practical application of CAD/CAM technologies, CNC machinery, automation, and industrial management systems.

13.1 Visit to Sands Precision Pvt. Ltd.

Guide: Mr. Dinesh Karanth

Sands Precision Pvt. Ltd. is a manufacturer of CNC machine tools, primarily CNC turning centres and machining centres, serving the precision manufacturing, automotive, aerospace, and general engineering sectors. The company employs approximately 30 personnel working in a single shift and manufactures nearly 100 CNC machines annually, with a machine utilisation rate of approximately 45%.

During the visit, we observed the complete lifecycle of a CNC machine — from design and component manufacturing, through sub-assembly and final assembly, to testing, quality inspection, and dispatch. Mr. Dinesh Karanth walked us through each stage of the process and explained how CAD/CAM software is used both in the design of the machines themselves and in programming the machines that manufacture the components.

Key observations at Sands Precision:

- CNC turning centres and vertical machining centres (VMCs) in various stages of assembly on the shop floor.
- The company uses CAD/CAM software for the internal design of machine components and for generating programs to machine precision parts on their own CNC machines.
- Quality inspection involved coordinate measuring machines (CMMs) and surface roughness testers to verify machined components before assembly.
- The assembly process involved precise fitting and alignment procedures to ensure the geometric accuracy of the final machine.
- Machine utilisation rate of 45% was explained as a result of the batch nature of custom CNC machine production — machines are not run continuously but in planned production batches.
- We learned that CNC machine companies like Sands Precision play a critical enabling role in the manufacturing ecosystem: the machines they produce are used by hundreds of downstream manufacturing companies.

Learning Outcome: The visit to Sands Precision provided a unique perspective on CNC technology — not just as a user of CNC machines, but as a manufacturer of them. It highlighted the depth of engineering knowledge required to design, build, and maintain CNC machine tools, and reinforced the importance of precision in every stage of manufacturing.

13.2 Visit to United Concord Industries

United Concord Industries is a large-scale manufacturer of pelletizing equipment used in various industrial processing applications, including plastics, food processing, and chemical industries. The company employs over 250 personnel and operates with a high machine utilisation rate of approximately 85%, reflecting continuous production.

Key observations at United Concord Industries:

- Large-scale fabrication and heavy machining operations involving lathes, milling machines, drilling machines, and CNC equipment working in an integrated production layout.
- The pelletizing machines manufactured here are large, complex assemblies requiring careful integration of fabricated structures, machined components, and purchased parts.
- Assembly and quality control operations were observed, with emphasis on dimensional verification and fit-up of mating components.
- The high machine utilisation rate (85%) was attributed to continuous, high-volume production orders and efficient production planning.
- CAD/CAM tools are used for the design of equipment and for planning CNC machining operations, with a strong emphasis on standardisation to enable rapid re-ordering and part replacement.
- The organisation of the production floor — with dedicated machining, sub-assembly, and final assembly areas — demonstrated the application of lean manufacturing and workflow management principles.

Learning Outcome: United Concord Industries demonstrated what high-volume industrial manufacturing looks like at scale. The experience highlighted the importance of production planning, standardisation, and integration of different manufacturing processes to achieve efficiency and quality at the industrial level.

13.3 Visit to Sree Ranga Engineering Works

Sree Ranga Engineering Works is a sheet metal fabrication and laser cutting specialist, employing approximately 50 personnel. The company provides laser cutting, sheet metal cutting, bending, and fabrication services to a wide range of industries. CAD software such as SolidWorks and AutoCAD is used for design preparation, while CNC-controlled laser machines ensure high precision and productivity.

Key observations at Sree Ranga Engineering Works:

- Advanced CNC laser cutting machines capable of cutting complex 2D profiles from sheet metal with high speed and precision — a direct application of CAD/CAM that mirrors our project.
- Sheet metal bending using CNC press brakes, with bend programs generated from CAD models — similar to the G-code-based programming we used for CNC milling.
- The use of SolidWorks for 3D design of sheet metal components and AutoCAD for 2D layout and nesting (optimising the arrangement of parts on a sheet to minimise waste).
- The nesting process demonstrated the economic importance of material utilisation — poorly nested parts waste expensive sheet metal, directly impacting profitability.
- A combination of automated CNC operations for cutting and manual operations for assembly, welding, and finishing — highlighting the complementary relationship between automation and skilled human labour.
- Quality checks were performed using measuring tapes, squares, and digital gauges to verify cut dimensions and bend angles.

Learning Outcome: Sree Ranga Engineering Works provided a clear view of how CAD/CAM is applied in the sheet metal industry. The parallels with our own CNC machining project were striking — digital design, toolpath/program generation, CNC execution, and quality verification follow the same fundamental sequence. The visit underscored the versatility of CAD/CAM principles across different manufacturing domains.

13.4 Consolidated Learnings from Industry Visits

Across all three industrial visits, several important cross-cutting observations and learnings were noted:

- Automation was adopted wherever economically justifiable, but skilled human operators remained essential for programming, supervision, machine maintenance, and quality assurance.
- CAD/CAM software is not just a design tool — it is a central part of the production system, connecting engineering design with manufacturing execution.
- The visits reinforced that manufacturing quality is the result of disciplined processes, not just good machines. Process planning, tooling selection, workholding, and quality inspection are all equally important.
- The industries demonstrated a range of scale — from a 30-person CNC machine builder to a 250-person heavy fabrication unit — illustrating that the same CAD/CAM principles apply across different scales of operation.
- There is a growing trend towards automation and robotics in all three industries visited. As labour costs rise and precision requirements increase, investment in CNC and automation technologies continues to grow.
- The students' interactions with engineers and supervisors highlighted the importance of combining technical skills (CAD/CAM, programming, machining) with practical problem-solving and communication abilities in the industrial environment.

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